

Machining Job Costing & Rate Calculation App

Functional Specification for Coding Agent Handoff

This document describes the complete calculation logic, data model, and design decisions for a workshop job-costing application. It is intended to brief a coding agent on exactly what the application must compute and how the data should be structured, ahead of a build prompt.

1. Purpose of the Application

The workshop performs machining work (Traub, lathe, drilling, grinding, rolling machines) on incoming components. For every new component/job, the shop needs to calculate a final quoted rate. The app must replace manual rate calculation with a structured, repeatable engine covering the full cost stack:

- Raw material cost
- Labour / machining operation cost
- Plating / surface treatment cost
- Packing & forwarding cost
- Transport cost
- Profit margin

Final Rate = Material Cost + Σ (Operation Costs) + Plating Cost + Packing/Forwarding + Transport, then + Margin%

2. Raw Material Cost Engine

The raw material cost is derived from the weight of the raw stock multiplied by its current market rate per kg. Weight is calculated from one universal master formula, driven by two configurable lookups: a Material Master (density) and a Shape Master (cross-section area formula).

Master Weight Formula

Weight (kg) = Cross-Section Area (mm²) × Length (mm) × Density (g/cm³) ÷ 1,000,000

Material Cost (Rs) = Weight (kg) × Current Rate per kg (Rs/kg)

2.1 Shape Master — Cross-Section Area Formulas

Shape	Required Inputs	Cross-Section Area Formula (mm ²)
Round solid bar	Diameter (d)	$(\pi/4) \times d^2$
Square solid bar	Side (s)	s^2
Hexagonal bar	Side / across-flats (s)	$2.598 \times s^2$
Round pipe / tube	Outer dia (OD), Inner dia (ID)	$(\pi/4) \times (OD^2 - ID^2)$
Flat / rectangular bar	Width (w), Thickness (t)	$w \times t$

Square/rect. tube (hollow)	Outer W×H, wall thickness	$(W \times H) - (w \times h)$ inner
Angle (L-section)	Leg a, Leg b, thickness (t)	$(a \times t) + (b - t) \times t$

Design note: implement this as a Shape Master table (shape name → formula/parameters), not as hardcoded per-shape branches, so new shapes can be added without code changes.

2.2 Material Master — Density Reference

Material	Typical Density (g/cm³)
Mild Steel (MS)	7.85
Stainless Steel (SS 304/316)	8.00
OHNS (Oil Hardened Non-Shrinking tool steel)	7.70 – 7.85
EN8 / EN9 alloy steel	7.85
Brass	8.40 – 8.70
Aluminium	2.70
Copper	8.90

Design note: this is an editable master table, not fixed constants — the shop owner must be able to add materials or correct density values.

2.3 Worked Example (validates the model)

Round MS bar, diameter 50 mm, length 100 mm: $\text{Area} = (\pi/4) \times 50^2 = 1963.5 \text{ mm}^2$. $\text{Weight} = 1963.5 \times 100 \times 7.85 \div 1,000,000 = 1.541 \text{ kg}$. This matches the shop's existing shortcut formula of $\text{dia}^2 \div 160,000 \times \text{length}$ (the constant 160,000 is this same formula pre-simplified and rounded for MS specifically). The app should use the full formula ($\text{area} \times \text{length} \times \text{density}$) so it generalizes correctly across all materials and shapes, rather than a single hardcoded constant.

3. Labour / Machining Operation Cost Engine

Machining operation cost is not a fixed engineering formula — it depends on the shop's own machines, tooling, and operator experience. The app must therefore treat this as a configurable Operation Master, fully custom (no pre-built starter list) so the shop owner defines every operation from scratch.

3.1 Operation Master — Data Model

Field	Description
Operation Name	e.g. Facing, Turning, Drilling, Grinding, Rolling (user-defined, free text)
Machine	Which machine performs it (Traub, Lathe, Drilling, Grinding, Rolling — user-defined)
Driving Parameter Type	The dimension that determines cost, e.g. Length, Diameter, Depth, Area, Number of passes, Per-
Rate per Unit	Rs per unit of the driving parameter (fully editable by the user)

3.2 Operation Cost Formula (kept intentionally simple)

Operation Cost (Rs) = Driving Parameter Value × Rate per Unit

No fixed setup cost is included per operation (explicit decision — keep it simple). A job's total labour cost is the sum of every operation added to that specific component:

Total Labour Cost (Rs) = Σ (Driving Parameter Value × Rate per Unit) for every operation on the job

Design note: the coding agent should NOT hardcode any list of operations, machines, or cutting-physics formulas (speed/feed/depth-of-cut calculations). The Operation Master is entirely user-populated and user-edited — the app only needs to support creating, editing, and applying these custom entries.

4. Plating / Surface Treatment Cost

Plating is charged per piece, based on the finished component's weight (after machining, i.e. after material removal) multiplied by the current plating rate per kg.

Finished Weight (kg) = Cross-Section Area (finished dims) × Finished Length × Density ÷ 1,000,000

Plating Cost (Rs) = Finished Weight (kg) × Current Plating Rate (Rs/kg)

Design note: the app must capture finished-piece dimensions separately from raw stock dimensions (they differ because material is removed during machining), using the same weight engine as Section 2.

5. Packing & Forwarding, Transport, and Margin

Cost Item	Basis (user selects per job)
Packing & Forwarding	Flat amount (Rs) OR per-kg rate (Rs/kg × finished weight)
Transport	Flat amount (Rs) OR per-kg rate (Rs/kg × finished weight)
Profit Margin	Percentage (%) applied to the running subtotal

6. Complete Calculation Flow (End-to-End)

Step	Calculation
1	Raw Material Cost = Raw Weight (via Shape + Material Master) × Current Material Rate/kg
2	Labour Cost = Σ (Driving Parameter × Rate per Unit) across all operations added to the job
3	Plating Cost = Finished Weight × Current Plating Rate/kg (if plating applies)
4	Subtotal = Raw Material Cost + Labour Cost + Plating Cost
5	Add Packing & Forwarding (flat or per-kg)
6	Add Transport (flat or per-kg)
7	Apply Profit Margin % on the running total to get Final Quoted Rate

7. Data Model Summary (Entities)

- **Material Master:** material name, density (g/cm³) — editable/extendable table
- **Shape Master:** shape name, required dimension fields, area formula — fixed set of standard shapes listed in Section 2.1
- **Operation Master:** operation name, machine, driving parameter type, rate per unit — fully custom, user-created from scratch
- **Job / Component Record:** raw material shape + dimensions + length + material type + material rate; list of operations applied with their parameter values; finished dimensions; plating rate (if applicable); packing/forwarding basis + value; transport basis + value; margin %

8. Open Decision — Needs Owner Input Before Build

Not yet decided: should this be a web app with saved/searchable job records (so past component quotes can be looked up and reused), or a single-session calculator with no persistence? This affects whether the coding agent needs to design a database/storage layer and job history/search UI, or just a stateless calculation form. This decision should be confirmed before the build prompt is finalized.

This document is a functional specification only. It defines what the application must calculate and how its data should be structured; it does not prescribe a technology stack or UI design, which will be defined in the build prompt.